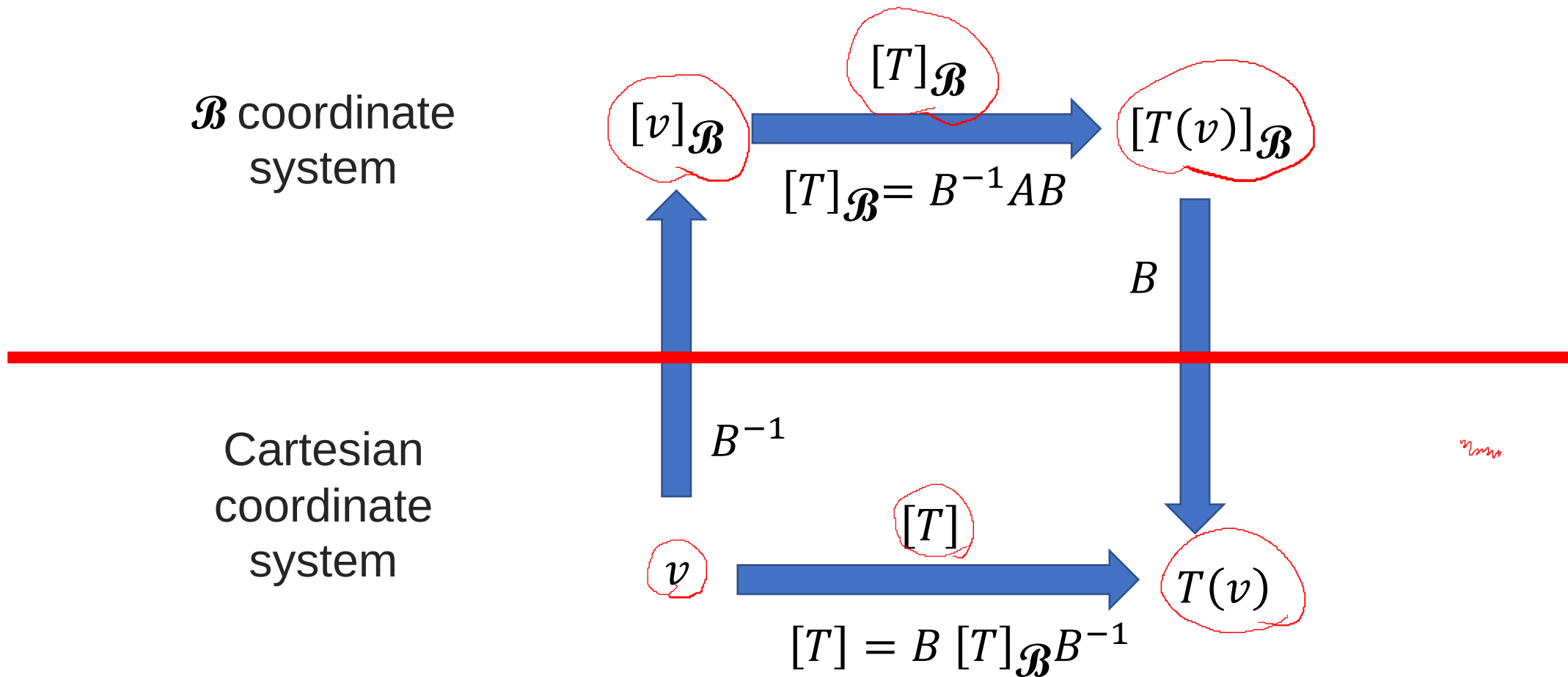


4-5

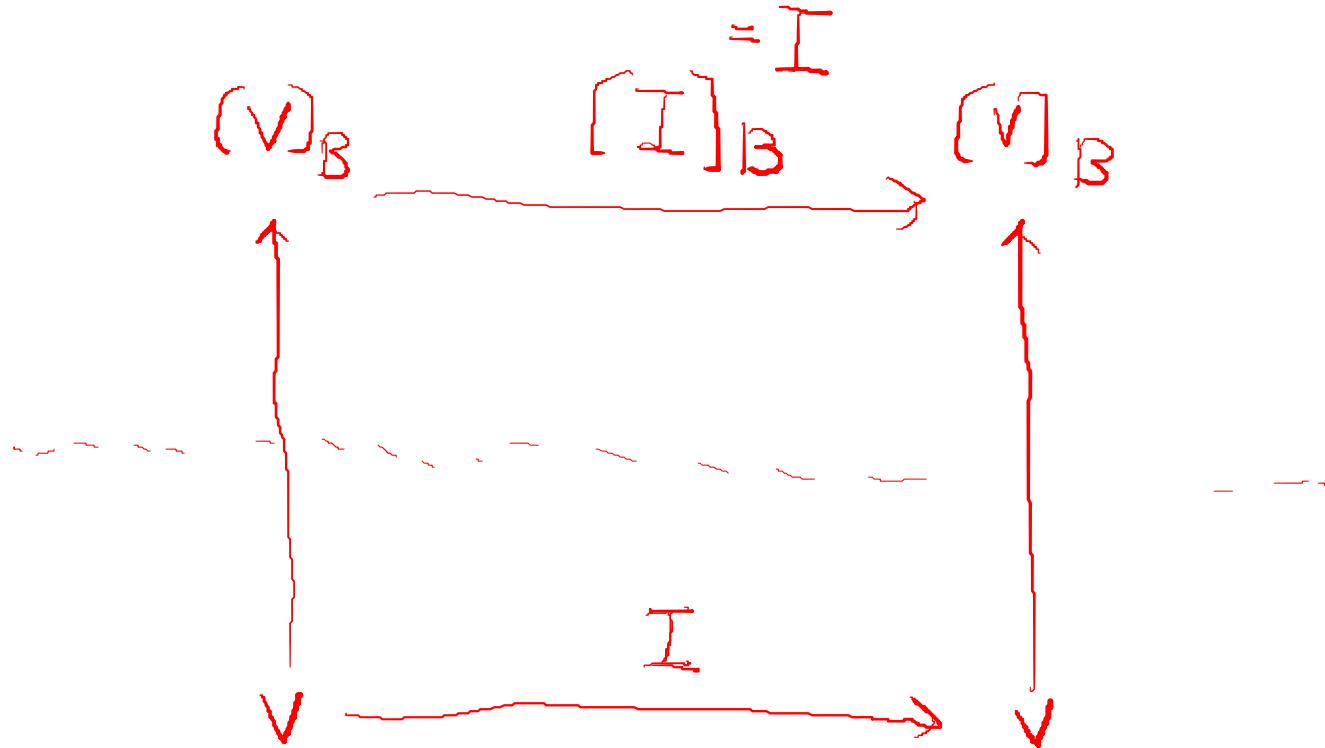


Review

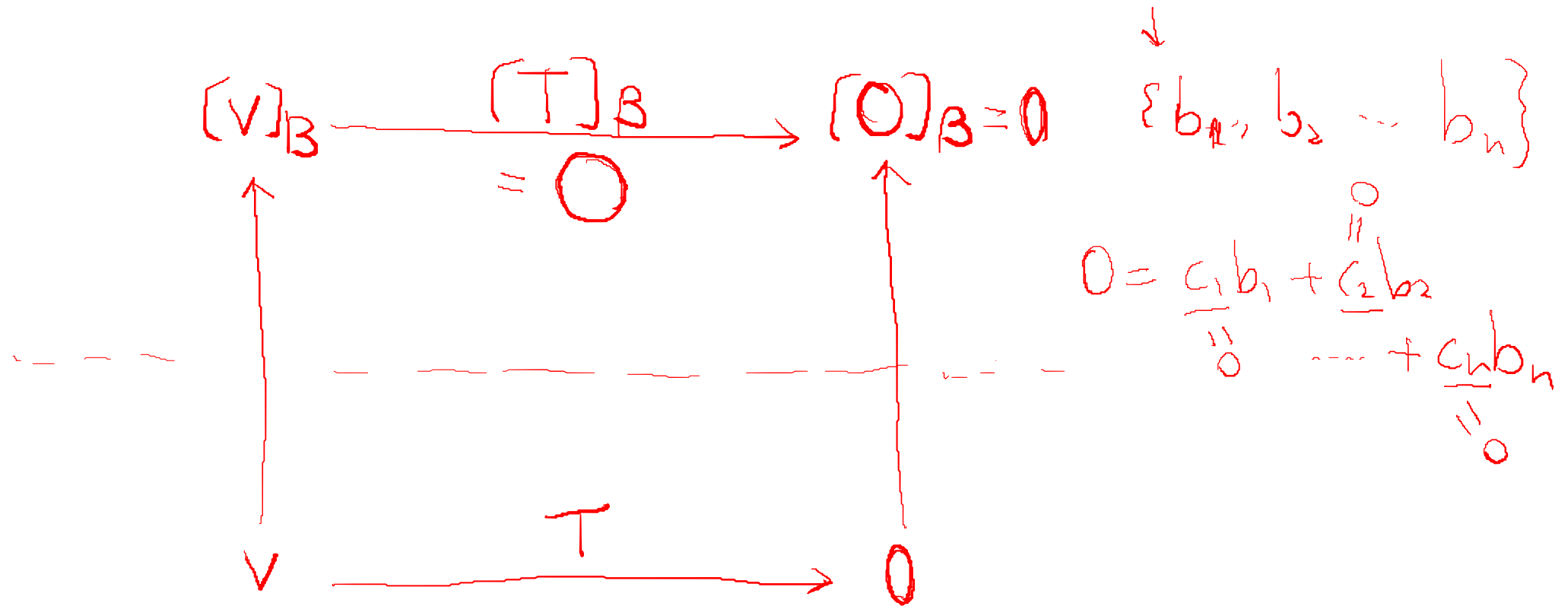
$$[v]_{\mathcal{B}}$$



63. Let I be the identity operator on R^n , and let B be any basis for R^n . Determine the matrix representation of I with respect to B .

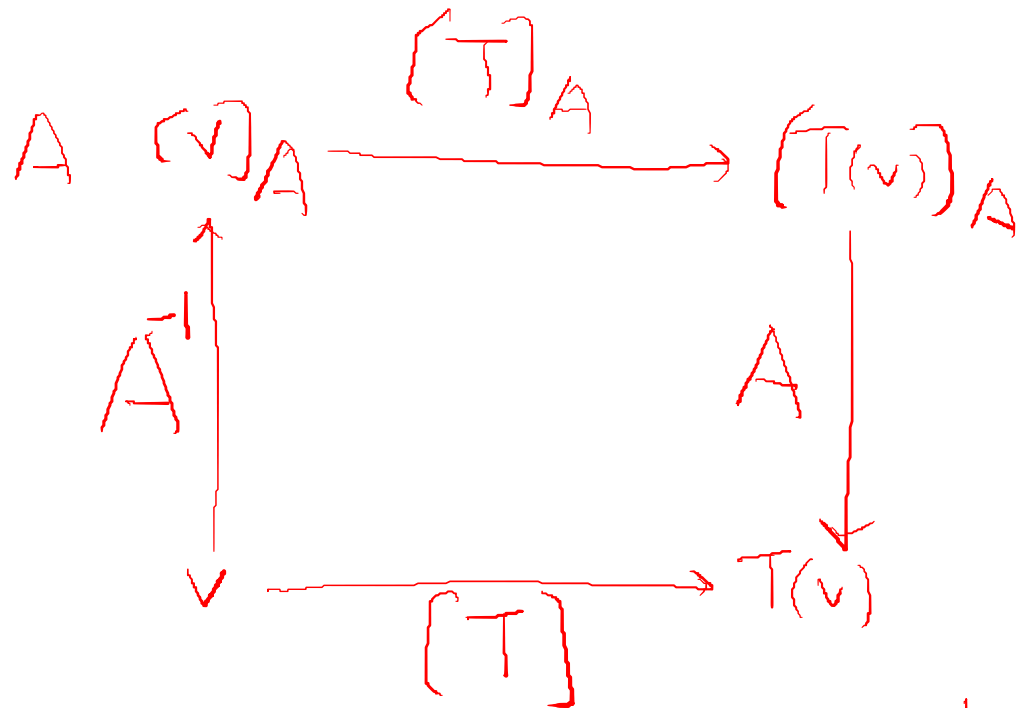


64. Let T be the zero operator on R^n , and let B be any basis for R^n . Determine the matrix representation of T with respect to B .



95. Let T be a linear operator on R^n , and let $\mathcal{A}$ and $\mathcal{B}$ be two bases for R^n . Prove that $[T]_{\mathcal{A}}$ and $[T]_{\mathcal{B}}$ are similar.

$$A = P B P^{-1}$$



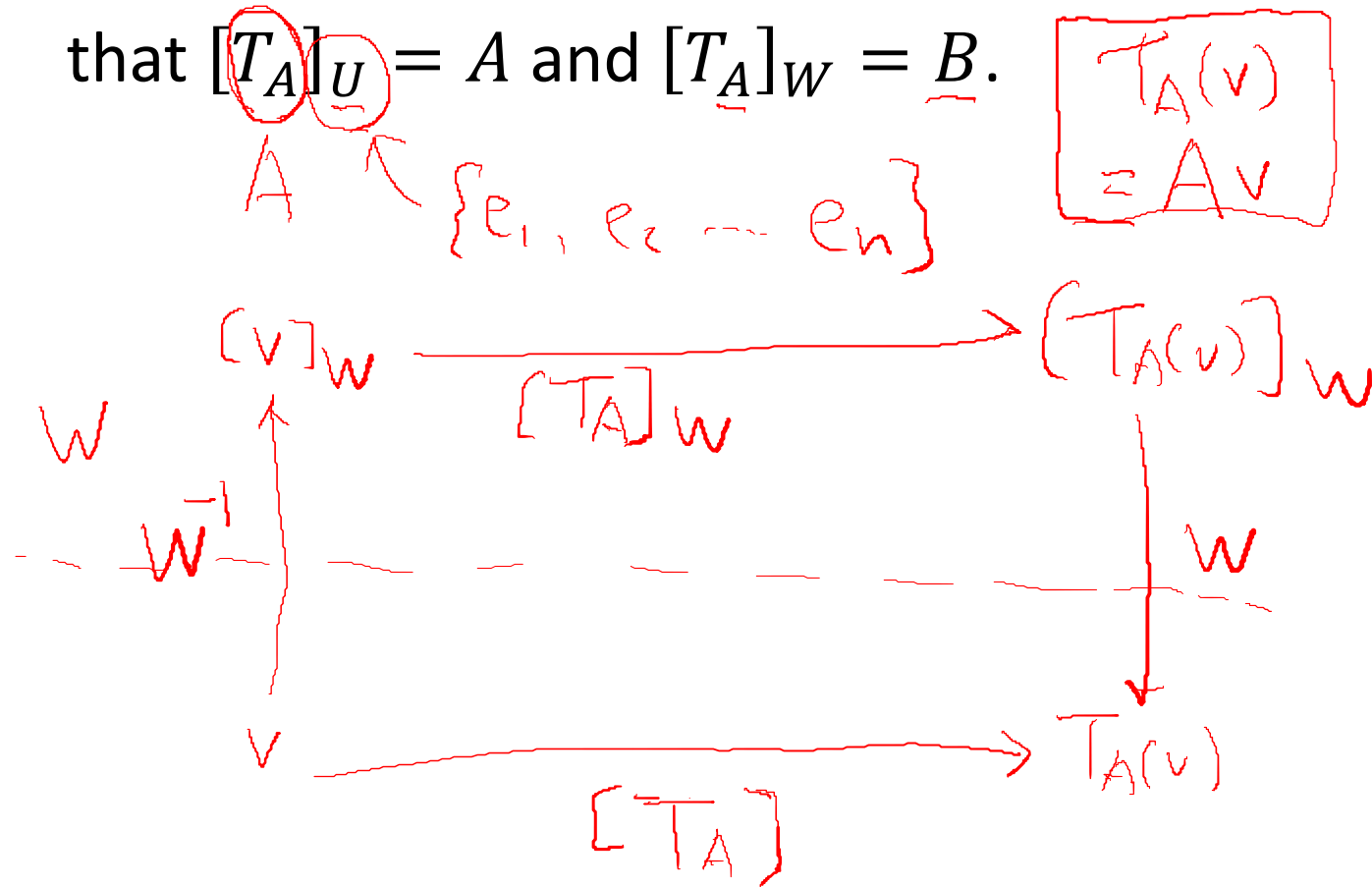
$$[T] = A [T]_A A^{-1}$$

$$A [T]_A A^{-1} = B [T]_B B^{-1}$$

$$[T]_A = \underbrace{A^{-1} B}_P [T]_B \underbrace{B^{-1} A}_{P^{-1}}$$

$$[T] = B [T]_B B^{-1}$$

96. Let A and B be similar matrices. Find bases $\underline{U}$ and $\underline{W}$ for R^n such that $[T_A]_{\underline{U}} = A$ and $[T_A]_{\underline{W}} = B$.



$$T_A(v) = Av$$

$$A = PBP^{-1}$$

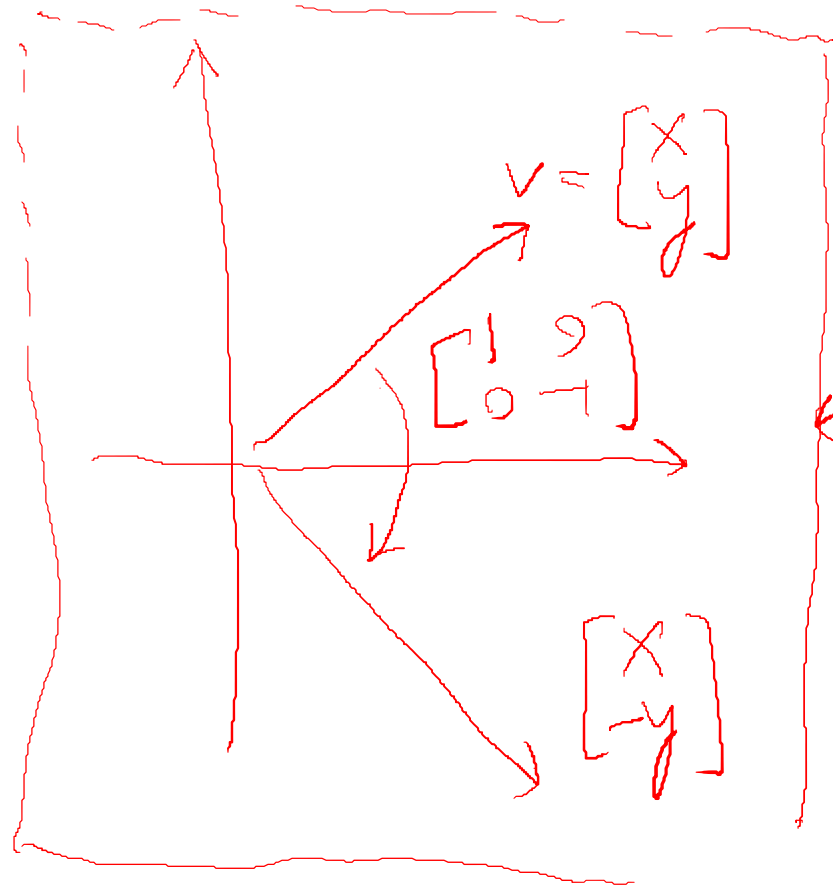
$$P^{-1}AP = B$$

$$= W[T_A]_W W^{-1}$$

$$[T_A]_W = W^{-1} [T_A] W = B$$

Similar matrices are matrix representations of the same linear operator.

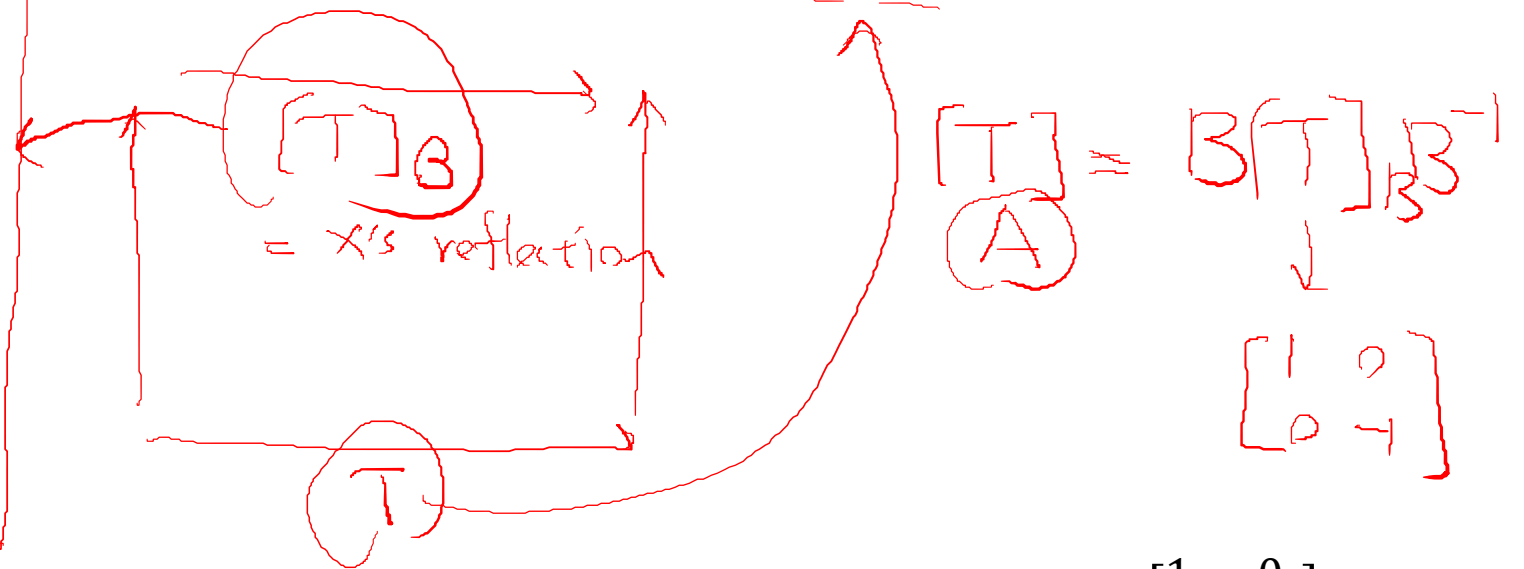
97. Show that if A is the standard matrix of a reflection of R^2 about a line, then $\det A = -1$. B?



$$\det A = \det \left(B \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} B^{-1} \right)$$

$$= \cancel{\det B} \det \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \cancel{\det B^{-1}} = -1$$

$\cancel{1/\det B}$



All the reflection matrices are similar to $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

81. Let U_W be the orthogonal projection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

$$B = \left\{ \overset{b_1}{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}, \overset{b_2}{\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}}, \overset{b_3}{\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}} \right\}$$

Note that the first two vectors in B lie in W , and the third vector is perpendicular to W . In general, we can apply a fact from geometry that the vector

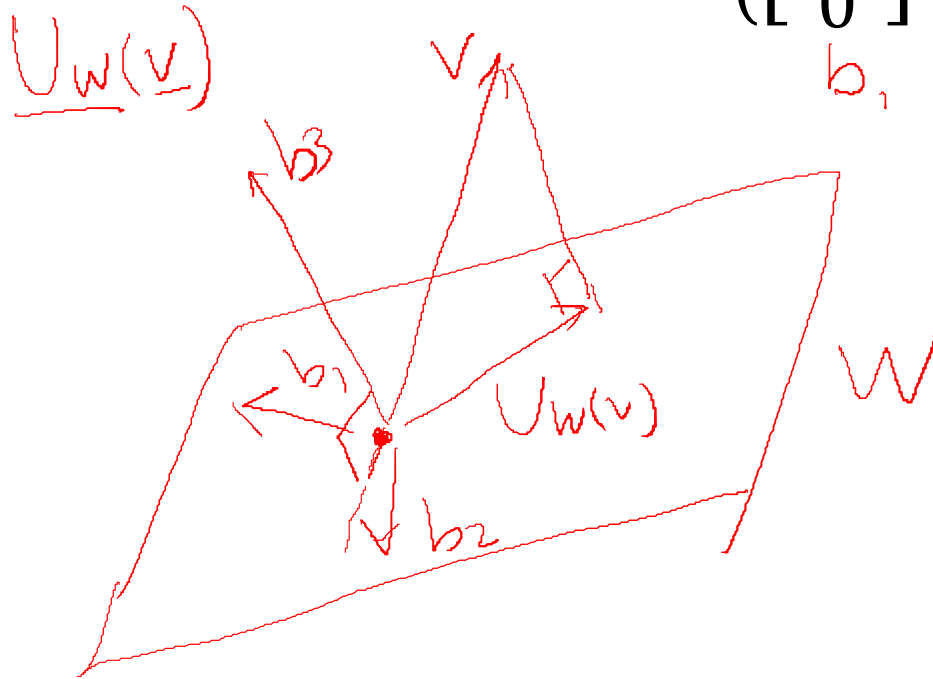
$$\begin{bmatrix} a \\ b \\ c \end{bmatrix},$$

whose components are the coefficients of the equation of the plane $ax + by + cz = d$, is normal (perpendicular) to the plane.

81. Let U_W be the orthogonal projection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

$$B = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}$$

$b_1 \quad b_2 \quad b_3$



81. Let U_W be the orthogonal projection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

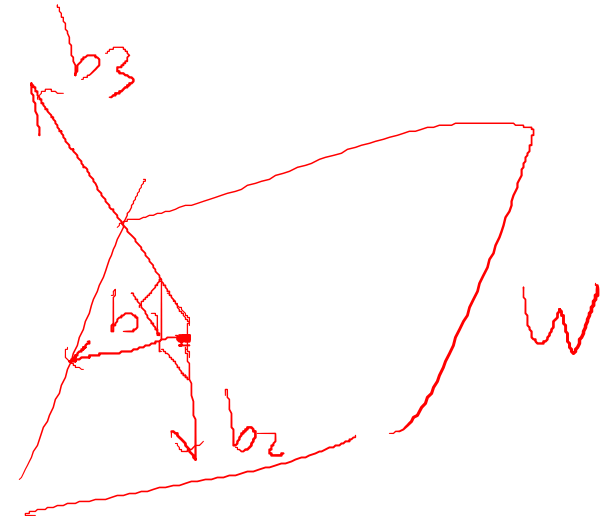
$$B = \left\{ \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{b_1}, \underbrace{\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}}_{b_2}, \underbrace{\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}}_{b_3} \right\}$$

(a) Find $U_W(\mathbf{v})$ for each vector $\mathbf{v}$ in B .

$$U_W(b_1) = b_1$$

$$U_W(b_2) = b_2$$

$$U_W(b_3) = \mathbf{0}$$



81. Let U_W be the orthogonal projection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

$$B = \left\{ \overset{b_1}{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}, \overset{b_2}{\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}}, \overset{b_3}{\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}} \right\}$$

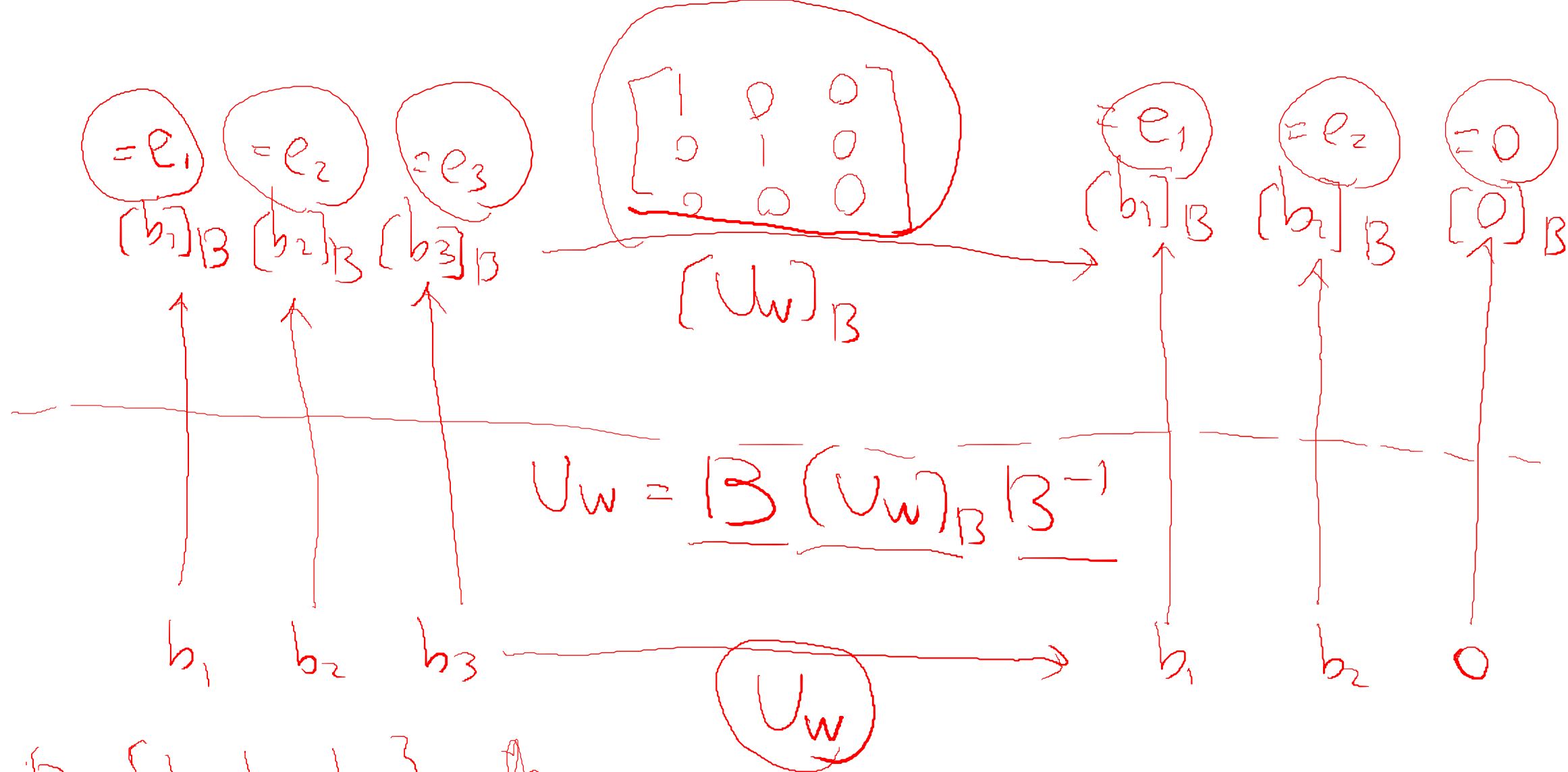
$$[b_1]_B [b_2]_B [b_3]_B$$

(b) Find $[U_W]_B$.

(c) Find the standard matrix of U_W .

$$[b_1]_B [b_2]_B [b_3]_B \xrightarrow{[U_W]_B}$$

$$\begin{matrix} & \uparrow \\ b_1 & b_2 & b_3 \end{matrix} \xrightarrow{U_W} \begin{matrix} b_1 & b_2 & 0 \end{matrix}$$



$$B = \{b_1, b_2, b_3\} \quad \text{th}$$

$$a_j = A e_j$$

73. Let T_W be the reflection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

$$B = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}$$

Note that the first two vectors in B lie in W , and the third vector is perpendicular to W . In general, we can apply a fact from geometry that the vector

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix},$$

whose components are the coefficients of the equation of the plane $ax + by + cz = d$, is normal (perpendicular) to the plane.

73. Let T_W be the reflection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

$$B = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}$$

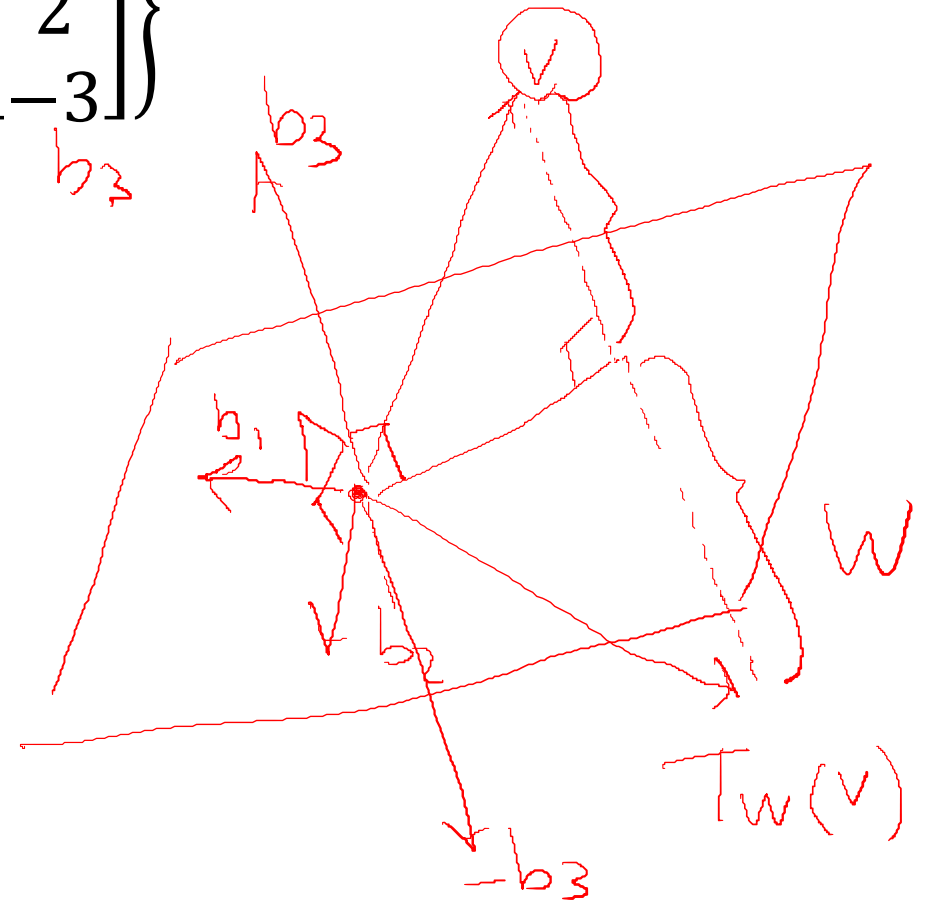
b_1 b_2 b_3

(a) Find $T_W(\mathbf{v})$ for each vector $\mathbf{v}$ in B .

$$T_W(b_1) = b_1$$

$$T_W(b_2) = b_2$$

$$T_W(b_3) = -b_3$$



73. Let T_W be the reflection of R^3 about the plane W in R^3 with equation $x + 2y - 3z = 0$, and let

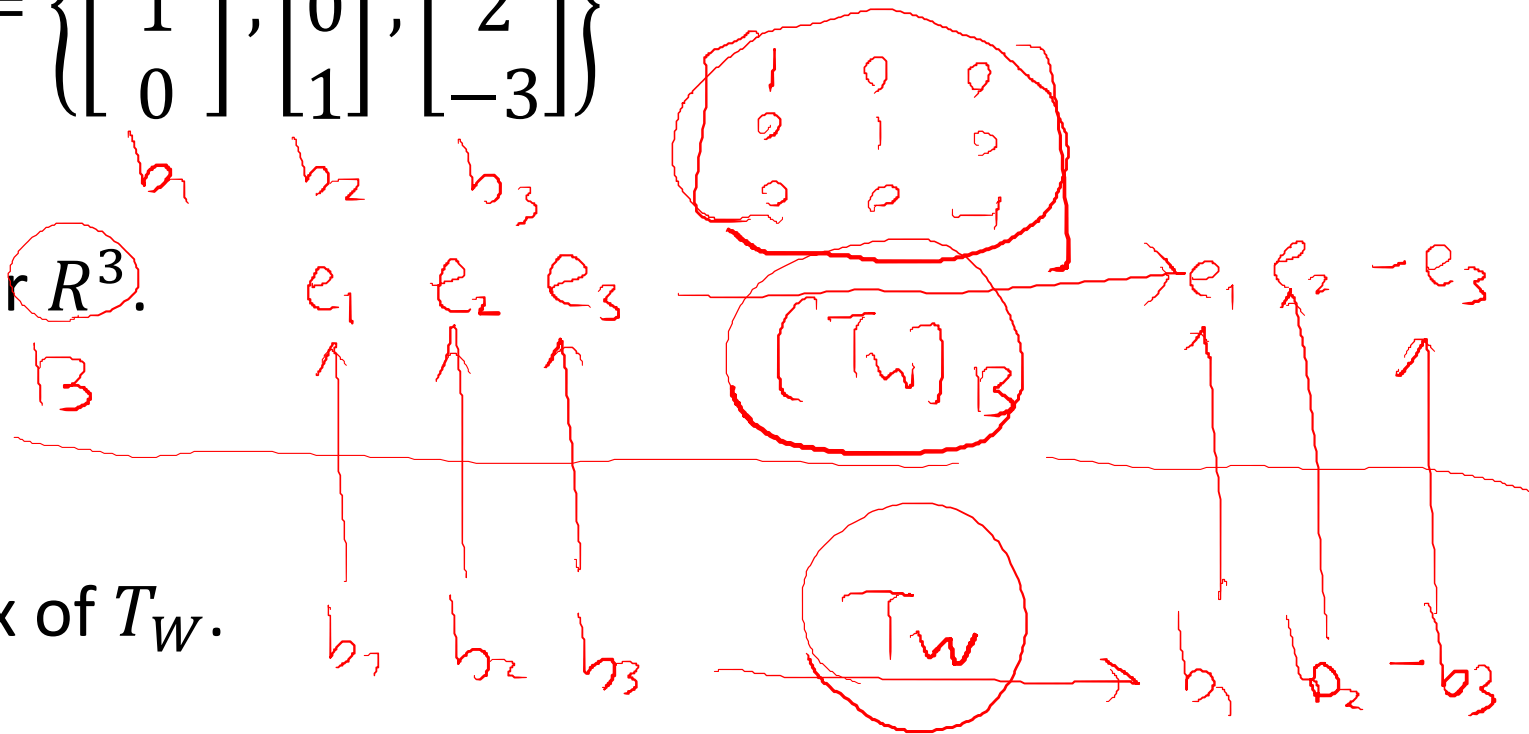
$$B = \left\{ \overset{\text{ind.}}{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \right\}$$

$b_1 \quad b_2 \quad b_3$

(b) Show that B is a basis for R^3 .

(c) Find $[T_W]_B$.

(d) Find the standard matrix of T_W .



98. Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for R^n , and let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ be (not necessarily distinct) vectors in R^n .

(a) Show that the matrix transformation T induced by CB^{-1} satisfies $T(\mathbf{b}_j) = \mathbf{c}_j$ for $j = 1, 2, \dots, n$.

98. Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for R^n , and let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ be (not necessarily distinct) vectors in R^n .

(b) Prove that the linear transformation in (a) is the unique linear transformation such that $T(\mathbf{b}_j) = \mathbf{c}_j$ for $j = 1, 2, \dots, n$.

Assume there is another linear transformation $U(\mathbf{b}_j) = \mathbf{c}_j$

98. Let $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for R^n , and let $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ be (not necessarily distinct) vectors in R^n .

(c) Extend these results to an arbitrary linear transformation $T: R^n \rightarrow R^m$

100 Let T be a linear operator on R^n and B be a basis for R^n . Prove the following results:

(a) For every vector $\boldsymbol{v}$ in R^n , $[T(\boldsymbol{v})]_B = [T]_B [\boldsymbol{v}]_B$

(b) If C is an $n \times n$ matrix such that $[T(\boldsymbol{v})]_B = C[\boldsymbol{v}]_B$ for every vector $\boldsymbol{v}$ in R^n , then $C = [T]_B$

How to know a matrix? Multiply by standard vector

Definition: Let $T: R^n \rightarrow R^m$ be a linear transformation, and let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ be bases for R^n and R^m , respectively. The matrix

$$[[T(\mathbf{b}_1)]_{\mathcal{C}} \quad [T(\mathbf{b}_2)]_{\mathcal{C}} \quad \dots \quad [T(\mathbf{b}_n)]_{\mathcal{C}}]$$

is called the matrix representation of T with respect to $\mathcal{B}$ and $\mathcal{C}$. It is denoted by $[T]_{\mathcal{B}}^{\mathcal{C}}$.

101. Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$ and $\mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$

(a) Prove that $\mathcal{B}$ and $\mathcal{C}$ are bases for R^3 and R^2 , respectively.

(b) Let $T: R^3 \rightarrow R^2$ be the linear transformation defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 - x_3 \\ x_1 - x_2 + 2x_3 \end{bmatrix} \quad \text{Find } [T]_{\mathcal{B}}^{\mathcal{C}}$$

102. Let $T: R^n \rightarrow R^m$ be a linear transformation, and let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ be bases for R^n and R^m , respectively. Let B and C be the matrices whose columns are the vectors in $\mathcal{B}$ and $\mathcal{C}$, respectively. Prove the following results:

(a) If A is the standard matrix of T , then $[T]_{\mathcal{B}}^{\mathcal{C}} = C^{-1}AB$

102. Let $T: R^n \rightarrow R^m$ be a linear transformation, and let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ be bases for R^n and R^m , respectively. Let B and C be the matrices whose columns are the vectors in $\mathcal{B}$ and $\mathcal{C}$, respectively. Prove the following results:

(b) If $U: R^n \rightarrow R^m$ is linear and s is any scalar, then

$$(i) [T + U]_{\mathcal{B}}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} + [U]_{\mathcal{B}}^{\mathcal{C}}$$

$$(ii) [sT]_{\mathcal{B}}^{\mathcal{C}} = s[T]_{\mathcal{B}}^{\mathcal{C}}$$

$$(iii) [T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[\mathbf{v}]_{\mathcal{B}}, \text{ for any vector } \mathbf{v} \text{ in } R^n.$$

102. Let $T: R^n \rightarrow R^m$ be a linear transformation, and let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ be bases for R^n and R^m , respectively. Let B and C be the matrices whose columns are the vectors in $\mathcal{B}$ and $\mathcal{C}$, respectively. Prove the following results:

(c) Let $U: R^m \rightarrow R^p$ be linear, and let $\mathcal{D}$ be a basis for R^p . Then

$$[UT]_{\mathcal{D}}^{\mathcal{D}} = [U]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}$$

102. Let $T: R^n \rightarrow R^m$ be a linear transformation, and let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_m\}$ be bases for R^n and R^m , respectively. Let B and C be the matrices whose columns are the vectors in $\mathcal{B}$ and $\mathcal{C}$, respectively. Prove the following results:

(d) Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\} \text{ and } \mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

Let $U: R^3 \rightarrow R^2$ be the linear transformation such that

$$[U]_{\mathcal{B}}^{\mathcal{C}} = \begin{bmatrix} 1 & -2 & 4 \\ 3 & -3 & 1 \end{bmatrix}$$

Find an explicit formula for $U(\mathbf{x})$

Review

51. Let B be a basis for R^n and T be an invertible linear operator on R^n .
Prove that $[T^{-1}]_B = ([T]_B)^{-1}$